EXPERIMENT 1: MATLAB CODE

Title: Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

AIM: To calculate the cutoff frequency and propagation constant of TE_{10} and TM_{11} modes in a WR-90 rectangular waveguide and plot their propagation characteristics.

MATLAB program

```
clc;
clear;
close all;
% Experiment 1: TE and TM Modes in Rectangular Waveguide
c = 3e8;              % Speed of light (m/s)
a = 22.86e-3;         % Waveguide width (m)
b = 10.16e-3;         % Waveguide height (m)
L = 100e-3;           % Waveguide length (m)
f = 10e9;             % Operating frequency (Hz)
% Cutoff frequencies
fc_TE10 = c/(2*a);
m = 1;
n = 1;
fc_TM11 = (c/2)*sqrt((m/a)^2 + (n/b)^2);
% Propagation constant
beta_TE10 = (2*pi*f/c)*sqrt(1-(fc_TE10/f)^2);
if f > fc_TM11
   beta_TM11 = (2*pi*f/c)*sqrt(1-(fc_TM11/f)^2);
else
   beta_TM11 = 0;
end
% Display calculated results
fprintf('Operating Frequency = %.2f GHz\n',f/1e9);
```

```
fprintf('TE10 Cutoff Frequency = %.3f GHz\n',fc_TE10/1e9);
fprintf('TM11 Cutoff Frequency = %.3f GHz\n',fc_TM11/1e9);
fprintf('TE10 Propagation Constant = %.2f rad/m\n',beta_TE10);
fprintf('TM11 Propagation Constant = %.2f rad/m\n',beta_TM11);
% Frequency sweep
freq = linspace(5e9,20e9,500);
beta1 = zeros(size(freq));
beta2 = zeros(size(freq));
for k = 1:length(freq)
    if freq(k) > fc_TE10
        beta1(k) = (2*pi*freq(k)/c)* ...
            sqrt(1-(fc_TE10/freq(k))^2);
    end
 if freq(k) > fc_TM11
        beta2(k) = (2*pi*freq(k)/c)* ...
            sqrt(1-(fc_TM11/freq(k))^2);
    end
end

% Plot propagation constant
figure;
plot(freq/1e9,beta1,'b','LineWidth',2);
hold on;
plot(freq/1e9,beta2,'r','LineWidth',2);
xline(fc_TE10/1e9,'--b','TE10 cutoff');
xline(fc_TM11/1e9,'--r','TM11 cutoff');
xlabel('Frequency (GHz)');
ylabel('Propagation Constant \beta (rad/m)');
title('Propagation Characteristics of TE10 and TM11 Modes');
```

```
legend('TE10','TM11','Location','northwest');
grid on;
hold off;
% Illustrative TE10 electric-field distribution
x = linspace(0,a,100);
y = linspace(0,b,50);
[X,Y] = meshgrid(x,y);
Ey = sin(pi*X/a);
figure;
surf(X*1000,Y*1000,Ey);
xlabel('Width x (mm)');
ylabel('Height y (mm)');
zlabel('Normalized E-field');
title('Illustrative TE10 Electric-Field Distribution');
shading interp;
colorbar;
```

Expected output (calculated)

Illustrative theoretical values; not measured HFSS results.

Parameter	**Expected value**
Operating frequency	10 GHz
TE_{10} cutoff frequency	6.562 GHz
TM_{11} cutoff frequency	16.42 GHz
TE_{10} propagation constant at 10 GHz	Approximately 158.2 rad/m
TM_{11} at 10 GHz	Below cutoff; no real propagating β

RESULT:

The cutoff frequencies and propagation characteristics of the TE_{10} and TM_{11} modes were calculated using MATLAB. At 10 GHz, the dominant TE_{10} mode propagates, whereas the TM_{11} mode is below cutoff.

OUTPUT:

```
==========================================
 MICROWAVE ENGINEERING - EXPERIMENT 1
==========================================
Operating Frequency = 10.00 GHz

CUTOFF FREQUENCIES
TEM  = 0 GHz (ideal two-conductor line)
TE10 = 7.500 GHz
TE20 = 15.000 GHz

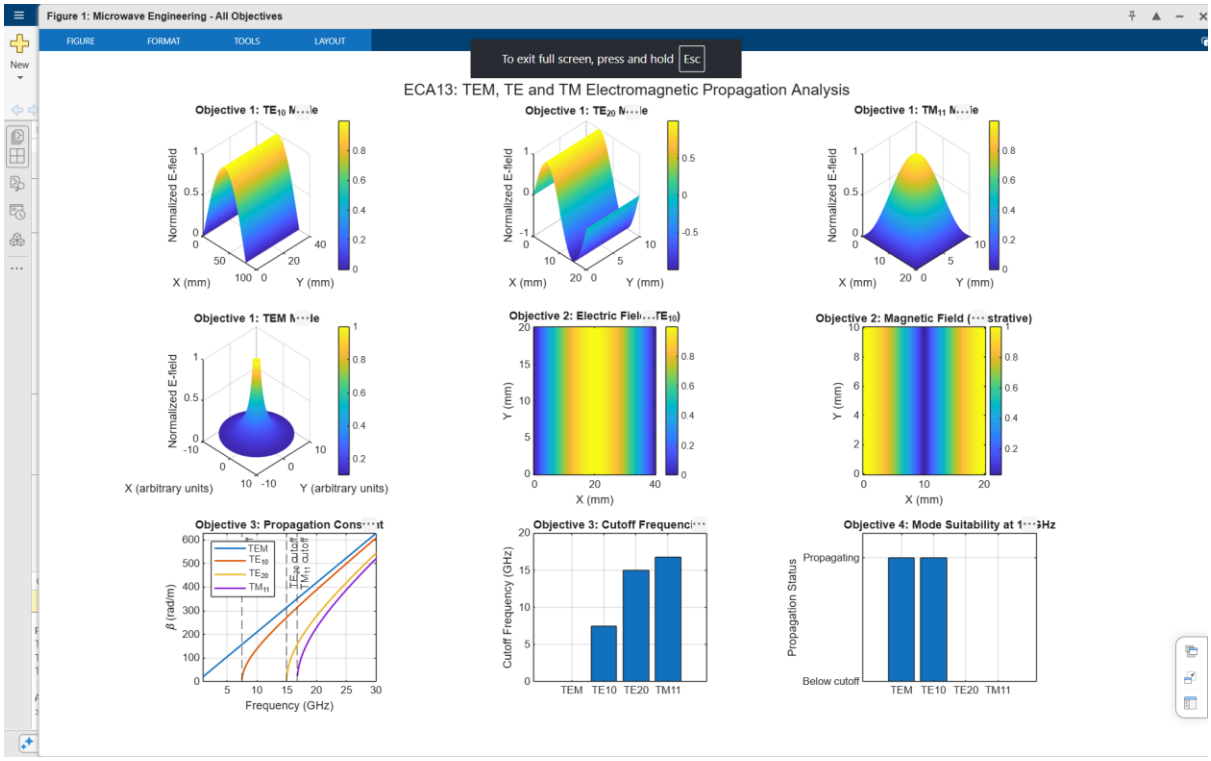
TM11 = 16.771 GHz

PROPAGATION CONSTANTS AT 10.00 GHz
TEM  beta = 209.440 rad/m
TE10 beta = 138.531 rad/m
TE20 is below cutoff at 10.00 GHz
TM11 is below cutoff at 10.00 GHz

PROPAGATION STATUS AT 10.00 GHz
TE10: Propagating
TE20: Below cutoff
TM11: Below cutoff

All four objectives are represented in one figure.

>> micro1
```


Figure 1: Microwave Engineering - All Objectives
FIGURE
FORMAT
TOOLS
LAYOUT
New
To exit full screen, press and hold Esc
ECA13: TEM, TE and TM Electromagnetic Propagation Analysis
Objective 1: TE10 Mode
Objective 1: TE20 Mode
Objective 1: TM11 Mode
Objective 1: TEM Mode
Objective 2: Electric Field (TE10)
Objective 2: Magnetic Field (Illustrative)
Objective 3: Propagation Constant
Objective 3: Cutoff Frequencies
Objective 4: Mode Suitability at 10 GHz
Normalized E-field
X (mm)
Y (mm)
X (arbitrary units)
Y (arbitrary units)
β (rad/m)
Frequency (GHz)
Cutoff Frequency (GHz)
Propagation Status
Propagating
Below cutoff
TEM
TE10
TE20
TM11